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Rethinking planning metrics: An analysis of common measurements of planning abilities

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ABSTRACT

Planning, or the ability to simulate and execute a sequence of steps toward a goal, is integral for success in a range of activities, from cooking a meal to developing a new software program. Indeed, robust planning abilities have been found to predict success in math for children and the maintenance of independence in older adults. Due to the significant real-world impact of developing and sustaining strong planning abilities over the lifetime, bolstering these skills through training could prove highly beneficial for a host of life outcomes. However, it remains unclear whether the measurement tasks currently used to assess planning abilities are addressing a consistent and cohesive cognitive construct. Although planning researchers have identified tasks they believe tap planning skills, these tasks have not been subjected to individual differences testing to assess the relationship between them. Therefore, the present study investigated the associations between three tasks used to measure planning in psychology and adjacent fields (the Tower of London task, the Zoo Map test, and the Traveling Salesperson Problems) in two samples; an original study and in a pre-registered replication. Contrary to their characterization as tasks designed to measure a skill under a common semantic label, the correlations between planning tasks found in this study were negligible and unstable between samples. The lack of reliable associations between tasks indicates that they are very likely not measuring the same underlying skill, and thus, should be viewed with caution when used to measure planning individually.

1. Introduction

Planning is the act of generating a set of steps to reach a particular goal (Oosterman et al., 2013). Given this, planning generally necessitates the ability to entertain different possible paths one could take, mentally simulate the potential outcomes of those paths, select the best path to the goal, and critically, adapt one's plan as new information arises. It follows then that planning skills are crucial for success in a variety of activities. This includes anticipating each subsequent step while solving a lengthy equation in mathematics, considering the possible reactions that may occur when adding different substances to a solution in a chemistry experiment, or even obtaining all necessary items during a trip to the grocery store (Morris & Ward, 2005). Consequently, children who excel at planning will tend to progress more successfully through math, science, and technology heavy courses while those who struggle with planning will be hindered (Cai et al., 2016). And older adults who maintain robust planning skills in later life will be able

to live independently longer than their peers with weaker planning skills (Hering et al., 2018; Kliegel et al., 2000; Lachman & Burack, 1993).

Given the massive real-world implications related to the development and maintenance of robust planning capabilities, interventions designed to enhance planning skills could offer a route to improved life outcomes for a variety of groups. Yet, before interventions on planning skills can be carried out, it is necessary to first establish valid measures of planning performance (i.e., that could be used to ascertain whether planning skills have been augmented as a result of some intervention). Unfortunately though, while many areas of both cognitive science and adjacent fields have their own sets of tasks that have traditionally been labeled as measures of planning, there is no consensus to date as to whether these different tasks measure the same construct. Indeed, some researchers have argued based upon theory that there may not be overall cohesion between tasks used to measure planning at all (see Burgess et al., 2005).

Over the decades of research on planning, researchers have taken

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different approaches to assessing planning skills. One broad category of tasks that have been developed are puzzle-style tasks. These tasks usually involve a set of highly constrained parameters and rules which can be solved via means-ends types analyses (i.e., identifying the initial problem state, defining the goal state, comparing the initial and goal states, and determining the actions or subgoals needed to connect the two). A classic and oft used puzzle-style task is the Tower of Hanoi (ToH) task which involves rearranging a set of differently sized discs on three pegs from a beginning arrangement to reach a specific goal arrangement (Simon, 1975). The key task constraints are that only the topmost disc on a given peg can be moved at a time, and a larger disc can never be placed on top of a smaller one (see Fig. 1 for an example problem). Other tasks that fit in this category include the highly related Tower of London (ToL) task (Shallice, 1982) as well as block-sliding puzzles such as the 8-puzzle used by O'Hara and Payne (1998), object displacement puzzles such as the Blocks Problem Solving Task (Waldron et al., 2006; Waldron et al., 2007), and even word problems with similar object-displacement mechanics such as the Missionaries-Cannibals task (Vernon & Strudensky, 1988).

A second broad category of tasks that have been given the label of planning tasks are real-world-planning-activity-emulation tasks. These include “toy” versions of running errands, planning an event, or navigating to various points via a map (Alderman et al., 2003; Hayes-Roth, 1980; Miotto & Morris, 1998; Morris et al., 2005; Schmitter-Edgecombe et al., 2012). Such tasks commonly require the participant to organize a list of objectives based on some criteria such as time efficiency, item priority, or due to the need for certain actions to be completed before others can occur. In a classic example of a real-world-planning-activity-emulation task, participants are given a list of many errands that can be completed by visiting various locations on a map of a fictional town or market area and are asked to create a practical plan outlining which errands to complete, when to complete them, and how to travel between each one (Hayes-Roth & Hayes-Roth, 1979). Other tasks that fit in this category include the Virtual Planning (VIP) task where the errands involve preparing for an overseas trip (Morris et al., 2005), and the Zoo Map task, where participants are asked to navigate the map of a fictional zoo visiting certain listed locations and following a set of movement rules including not retracing one's steps (Ballhausen et al., 2017; Oosterman et al., 2013). Importantly, while real-world-planning-activity-emulation tasks occur in many different settings and with a variety of constraints and goals, they all attempt to emulate planning activities that individuals might face in daily life in an effort to measure planning naturalistically.

A third broad category of tasks are those that have been commonly used to compare the planning capacities of computer algorithms and human participants. Within this category are various types of board games such as Chess and Go (van Opheusden & Ma, 2019) and simple route efficiency problems such as the Traveling Salesperson problems (Lihoreau et al., 2022; Macgregor & Ormerod, 1996; Vickers et al., 2001). These tasks differ from the other two categories of planning tasks in that they increase the space of possible move choices exponentially from the basic rule adherence task model. This allows researchers to approach the limits of computational planning capacity while remaining within the abilities of human participants to plan effectively. For

instance, Traveling Salesperson problems require participants to plan the shortest possible route to visit a set of spatial locations and to return to the starting location without crossing a previously drawn path, or visiting a previously visited location (Lihoreau et al., 2022). The problems can become more or less difficult to plan efficiently as the number of “internal nodes” (or locations that require the planner to journey into the center of the shape made by the external location dots) increases (Vickers et al., 2001).

Over the past several decades, researchers have argued for and against examples from each of these categories as effective measures of planning (see Burgess et al., 2005 and Berg et al., 2006 for discussion of these concerns). In particular, the ToH and related puzzle tasks have been suggested to focus on the measurement of what is referred to as “look-ahead,” which is defined as “the function of working out in one's mind every stage between the present state and some desired outcome” (Burgess et al., 2005, pg. 201), as a surrogate for planning in general. However, some researchers have asserted that simply looking ahead in time toward a concrete and well-defined goal does not capture the true complexity of planning in the contexts in which it is most commonly employed in situ (Shallice & Burgess, 1991). Researchers having this latter point of view have suggested the use of more ecologically-inspired measurement paradigms such as tasks that emulate real-life planning situations. Errand, event, meal, route, and other real-world emulating planning tasks are also though not without criticism. In particular, the variation in instructions, goals, and outcome measures (i.e., planning time, efficient sequence points, overall distance efficiency, etc.), may produce overly complex and somewhat intractable outcomes resulting in the inability to compare planning performance across studies using slightly different methodologies. Finally, similar to puzzle-style tasks, it is unclear whether planning tasks used to compare computer algorithms and human performance truly captures the qualities of decisions that planners often employ when considering a route or steps to solve a problem due to the lack of contextual information present in these task spaces.

Although overviews of the field have sometimes suggested a potential (problematic) lack of cohesion across the tasks in terms of what is truly being measured (Burgess et al., 2005; Davies, 2005; Phillips et al., 2005), there is currently a dearth of empirical evidence speaking to this point. This is a clear issue, as in many areas of psychology, the question of whether tasks that are given the same shared semantic label are adequately aligned with the same cognitive constructs that they refer to is prevalent. For instance, the shared label of “attention task” can refer to what we now think of as measuring a host of relatively uncorrelated constructs (e.g., “sustained attention” tasks such as the Mackworth clock task or various continuous-performance tasks (CPT) are not measuring the same construct as “selective attention” tasks such as visual search tasks). The use of common semantic labels for unrelated constructs could then present a significant obstacle for knock-on research, such as intervention studies.

There are several types of quantitative analysis that can be used to clarify whether a set of tasks truly measures a common underlying construct, but one frequent first step is to explore the bivariate correlations between tasks that are given a common semantic label. In the case of planning, it is clear the tasks using a common semantic label are coming at the issue from quite different angles. What is less clear though is what this in turn means in terms of the underlying construct that is measured.

The present study therefore sought to examine the strength of the associations between performance on three commonly used planning measurement tasks (one from each of the broad categories of tasks described above): the Tower of London task, the Zoo Map test, and the Traveling Salesperson problems. Ten additional cognitive skill and personality metrics were included in an effort to better understand the skills and traits associated with planning task performance (i.e., which cognitive skills or behavioral metrics might be associated with planning performance on these selected tasks). If the planning tasks are indeed



Fig. 1. Example of the Tower of Hanoi (ToH) task with Starting Configuration and Goal Configuration states depicted. Participants must move the disks between the three pegs in accordance with the set movement rules with the aim of replicating the “Goal Configuration” on the set of pegs provided as the “Starting Configuration.” In this example, five moves are necessary to solve the puzzle.

measuring a cohesive construct of planning, we would expect to see (1) strong correlations between these tasks and (2) common patterns of correlations between performance on the three tasks and the more basic measures of cognitive performance and personality. However, due to the differences in the measurement tasks used to assess planning abilities, and the suggestions from previous literature about the implications of these differences in task structure and loading (see Scholnick et al., 1997, and see Burgess et al., 2005), we hypothesized that such expectations would not be borne out. Instead, we hypothesized that the correlations between the included measures of planning would not be large enough to surpass the Bayesian *t*-test alternative hypothesis threshold (Rouder et al., 2009), indicating that measures given the label of “planning task” may not be consistently measuring the same cognitive construct.

2. Methods

The present study was a correlational study in which participants were tested on a battery of 13 tasks and survey metrics including a brief demographic survey (see Materials section for details on these measures). The participants included in the proceeding analyses are from two separate samples of participants, one original sample (Study 1), and one pre-registered study (Study 2) designed to replicate the findings from the original sample.

2.1. Participants

Altogether, Study 1 and Study 2 included a total of 174 participants. Study 1 included 67 participants, while Study 2 included 107 new participants, following exclusions (with no overlap between participants in the two samples; see Supplementary Material Table S1 for details on excluded participants: *N* = 2 for Study 1, and *N* = 1 for Study 2). Further information about exclusion procedures can be found in the *Exclusion of data* section below.

All participants were undergraduate students at the University of Wisconsin-Madison, which is a large Midwestern public university in the United States. Participants were recruited from the psychology department research participant pool, were compensated with course credit, had normal or corrected to normal vision, no major neurological deficits, and were 18 years old or older. Recruitment took place over the course of three semesters (approximately one full calendar year). See Table 1 for more demographic details.

2.1.1. Study 1 - original study

The original study (Study 1) consisted of two sessions which took place on two separate days. The maximum number of days between sessions was restricted to 10 days. On day one, participants were

measured on a battery of 13 measures (three planning measures plus 10 additional metrics), which took about two hours on average to complete. On day two, participants were asked to play five online mini video games and fill out a questionnaire detailing their previous experience playing video games (v.2021, Green et al., 2017). However, only the first session (task battery) is considered as part of the current work. The order of cognitive measures experienced by participants was counterbalanced such that roughly half of participants were assigned to one order of the tasks while the rest of the participants were assigned to experience the tasks in the reverse order. Participants were assigned a task order alternating between the two possible orders depending on when the participant arrived to participate in the study (i.e., the first participant was assigned to order one while the second participant was assigned to order two and so on). This counterbalancing was implemented in order to account for potential order effects.

2.1.2. Study 2 - replication study

A pre-registered replication study (Study 2) was conducted in which only the battery of measurement tasks was run (i.e., just those tasks completed during Session 1 of the original Study 1) in a manner identical to that described above for Study 1.

2.2. Materials

Thirteen cognitive measures were included in the task battery, three of which have been used to measure planning in prior literature. The three planning measures were the Tower of London task (Phillips et al., 2001), the Zoo Map test (Oosterman et al., 2013), and a set of Traveling Salesperson problems as used by Macgregor and Ormerod (1996).

2.2.1. The Tower of London

The Tower of London task is commonly used in cognitive psychology to measure planning skills (Davies, 2005; Phillips et al., 2001). In the task, similar to the Tower of Hanoi, participants are asked to move the three colored balls on and off the three different length pegs to match a target pattern within a certain number of moves. Participants are given three attempts to solve each problem and are awarded a score based on how many attempts they took to solve the problem (e.g., arriving at the correct solution on the 3rd attempt would award the participant a score of one, while solving the problem on the first attempt would award the participant a score of three). Thus, the fewer attempts that are needed before successful completion of each problem, the higher the score (Krikorian et al., 1994). The dependent variable for this task was the total score, which was the sum of the 12 problem scores, with the highest possible score being 36, as per the standard for this version of this task (Shallice, 1982).

2.2.2. The Zoo Map test

The Zoo Map test is commonly used to measure planning capabilities both in children and in clinical settings with adult neurological disorder patients. The test asks participants to plan out a route on the map of a zoo to visit a set of predefined locations while following movement rules that constrain the possible route choices (Ballhausen et al., 2017; Oosterman et al., 2013). There are eight total locations to visit, and four possible correct sequences of locations visited.

The dependent variable used for this task was the number of locations visited in one of the four possible correct orders. This dependent variable was selected because it provided a better score spread across our sample of healthy young adult participants than a difference metric between the planning task and a purely motoric version without the planning component. The latter is often used in clinical or child populations who may struggle with motoric output, which was not the case in our sample. The maximum score was eight if a participant visited all eight locations in one of the correct sequences, and the minimum score was zero if none of the locations were visited in an acceptable order based on movement constraints.

Table 1
Sample Demographic Information. Demographic information for both original and replication samples.

	N	Study 1 Sample	Study 2 Sample
Total Sample (without missing data)	174	67	107
Age	169		
Mean (SD)		18.6 (1.3)	18.7 (0.8)
Min, Max		18.0, 28.0	18.0, 22.0
Gender	170		
Female		35 (54 %)	43 (41 %)
Male		30 (46 %)	62 (59 %)
Education Completed	170		
Associate degree (2 year college/ junior college)		2 (3.1 %)	3 (2.9 %)
Bachelor's degree		1 (1.5 %)	1 (1 %)
High school degree or equivalent		62 (95 %)	101 (96 %)

Note: Five participants from the full sample did not provide some or all demographic information.

2.2.3. The Traveling Salesperson problems

The Traveling Salesperson Problems are commonly used in computer science to measure perceptual route efficiency in machine learning algorithms but are also used with human participants to measure the efficiency of route planning (Lihoreau et al., 2022; Macgregor & Ormerod, 1996; Vickers et al., 2001). For each of the included problems, participants were asked to draw the shortest route which visits each of the provided coordinate points without returning to a previously visited point and without crossing over a previously drawn path. Fourteen problems were included in the current study seven of which had 10 points and seven of which included 20 points and varying numbers of internal nodes which modulated difficulty (Macgregor & Ormerod, 1996).

The dependent variable was the distance traveled between points for each problem with penalties added for rule breaks (see Table 2 for exact variable calculation procedure). Although prior literature has not utilized distance penalties when administering this task in the past, this was necessary in the current dataset due to the numerous instances of rule breaking behavior in both samples. The assessment of penalties resulted in adjusted scores that ensured that better scores represented what would be considered more successful performance as opposed to instances where the task was simply completed incorrectly. For example, skipping a point in a problem may reduce the overall distance score for that problem, however it also reflects an incorrect completion of the problem, which requires visiting every point in the shape. After penalties were assessed, adjusted distance scores were averaged across the 14 problems completed by each participant. Longer distances indicated lower route efficiency, and thus, less successful planning.

2.2.4. Additional cognitive and personality measures

A variety of cognitive, behavioral, and personality metrics were included in the suite of measurements in addition to the measures of planning to provide more detailed results.

2.2.4.1. Cognitive skill metrics. Several of the remaining ten measures were tasks that tapped a variety of different cognitive skills with the aim of better understanding the cognitive skills associated with planning abilities.

2.2.4.1.1. Fluid intelligence. The University of California Matrix Reasoning Task (UCMRT) was included as a measurement of fluid intelligence. The UCMRT is a matrix reasoning task in which participants must select the shape that correctly completes the pattern in a 3×3 matrix of shapes with different colors and sizes (Pahor et al., 2019). Participants were asked to complete as many problems as possible within 10 min and the dependent variable included was the percent correct out of 23 total problems, including unattempted problems.

2.2.4.1.2. Spatial cognition. The Mental Paper Folding task is a

measurement of spatial reasoning and mental imagery. In the task, participants view a figure showing a sequence of folds undergone by a square piece of paper and then shown the location of a hole that has been punched in a particular location on the folded piece of paper. They are then asked to indicate the correct pattern of holes the paper would have if it were to be unfolded by selecting an answer from a set of possible solution options (Shepard & Feng, 1972). Participants were asked to complete as many problems as possible within three minutes, and the dependent variable was the percent correct out of 10 total problems, including unattempted problems.

2.2.4.1.3. Sustained attention. The gradual-onset continuous performance task (gradCPT) was included as a measurement of sustained attention. Stimuli were grayscale images of cities and mountains. On each trial (800 ms), one image gradually transitioned to the next using linear pixel-by-pixel interpolation. Participants were instructed to respond via button press to city scenes (90 %) and to withhold response to mountains (10 %; Rosenberg et al., 2013). Participants saw 500 images during this task and the dependent variable was d prime.

2.2.4.1.4. Working memory. The Backward Corsi Block Tapping Span task was included as a measure of working memory. In the task, participants were asked to view a set of eight blocks, a subset of which were highlighted in succession, and respond by clicking the highlighted blocks in the reverse order of the order they were highlighted in (Kessels et al., 2008). Participants received two attempts for each sequence length of highlighted blocks beginning with a two block sequence and increasing in sequence length until the participant could no longer recreate the reverse sequence after two attempts. The maximum number of block sequences correctly replicated (also called the block span) was used as the dependent variable for this study.

2.2.4.1.5. Cognitive flexibility. The Predictable Task Switching task was used to measure cognitive flexibility and asked participants to switch between two different tasks, a digit discrimination task and a letter discrimination task depending on the location of the pair of character stimuli presented within a 2×2 matrix. The stimulus pair moved clockwise from one quadrant of the matrix to the next and participants were asked to respond with whether the digit in the stimulus pair was even or odd in the digit task and whether the letter was a vowel or a consonant in the letter task (Rogers & Monsell, 1995). The dependent variable used in this study was difference in average reaction time between trials when the task on the previous trial was a different task then the task on the current trial (called switch trials) and trials when the previous trial task and the current trial task were the same (called non-switch trials). For the analysis, incorrect trials and post error trials were removed.

2.2.4.1.6. Cognitive placekeeping. The UNRAVEL task was included as a measure of placekeeping with task interruptions. In this task, participants were asked to remember their place in a sequence of binary choice tasks based on presented stimuli after experiencing an interruption in the form of another task which required mental focus to accomplish (Altmann & Trafton, 2015). The dependent variable was the reaction time and the accuracy of responses averaged across trials one trial after an interruption had occurred.

2.2.4.1.7. Speed of processing. As a measurement of processing speed, a task in which participants made judgments about the direction an arrow was pointing on the screen as quickly as possible via a button press was included (Dale et al., 2021). The dependent variable was average trial reaction time across all trials after incorrect trials and post error trials were removed.

2.2.4.2. Personality and individual trait metrics. The Shifting Shapes task was included as a measure of creativity and asked participants to first create as many interesting shapes as possible in five minutes with a set of ten connected squares and then save the five most interesting shapes from the set of shapes the participant had made (Hart et al., 2017). The dependent variables for this study were the number of unique shapes

Table 2
Traveling Salesperson Penalty Procedure.

Rule-Breaking Behavior	Penalty Description	Reasoning
Crossed line	For each shape, the average segment length was calculated from participants who followed all task rules. This average segment length was then doubled, and the resulting distance was added to the distance calculation for that shape of any participant who broke a rule for that shape.	Penalties were assessed in this way to account for incorrect task actions that would otherwise result in a shorter distance (and thus better score) than the scores of those who completed the task correctly.
Revisited point		
Skipped point		

Score penalty assessment procedure for the Traveling Salesperson task. For each instance of a rule-break in a given problem, one penalty length was added to the problem distance. Thus, a problem in which two rules were broken, or one rule broken twice, would be assessed $2 \times$ the penalty distance for that problem.

and the number of total shapes generated by the participant. The Vocabulary subtest of the Shipley Institute of Living Scale was included as a measure of crystallized intelligence. This task required participants to select the closest synonym out of a set of options for 43 common and uncommon words (Shipley, 1940). The dependent variable for this task was percent correct out of 43 words, including unanswered questions. Finally, the ten question version of the Big 5 Personality Inventory was included, which characterized each participant on the five axes of personality; agreeableness, conscientiousness, neuroticism, extraversion, and openness. One dependent variable was extracted for each trait, which was the composite score of one forward coded and one reverse coded item assessing each trait in the survey (Rammstedt & John, 2007). The abbreviated BFI-10 scale was used in place of the full 44-item version, as we aimed to minimize participant burden given the already substantial length of the study. Research has shown that the BFI-10 serves as a reliable substitute for the full scale, with an average correlation of 0.83 between the two versions across three samples (using Fisher's r -to- Z transformation). Correlations for individual subcategories ranged from 0.74 to 0.90 (Rammstedt & John, 2007).

2.3. Analysis plan

The replication data collection and subsequent analyses were pre-registered on OSF. As it is an important first step in exploring the cohesiveness of a semantic grouping such as planning, bivariate correlations were computed between the three planning measurement tasks and between these three tasks and the additional metrics collected in the study to explore further possible relationships between the tasks of interest and related skills and traits. All data analysis was completed in R (v4.4.0, R Core Team, 2024).

2.3.1. Exclusion of data

Per our pre-registration, data from individual participants in a given task was removed if they clearly demonstrated noncompliance on that task. If a participant needed to be removed from more than one task for non-compliance, they were removed from the full dataset. Participants whose data was missing due to technical issues or researcher error were retained in the remainder of the data set. In total, three participants were removed from the full data set, two for issues of noncompliance (one from each sample) and one additional participant from the original sample due to reporting their age as 17, making them ineligible to participate. Individual task or trial data was removed on a case by case basis in accordance with pre-registered guidelines. Missing individual task data in this way did not lead to the full exclusion of any participant from the sample (see Supplementary Material Table S1 for full description of data exclusions).

3. Results

In accordance with our pre-registered plans for statistical analyses, Spearman's rank correlations between all dependent variables from the three planning measurement tasks were computed to explore the relationship between performance on each individual task (Spearman, 1904). Dependent variables where a lower value represented better performance were reversed to align with other measures. Reversed scores were computed by multiplying the original score by -1 . As such, for all measures below with a clear directionality, larger values represent better performance. Additionally, we assessed the Bayes Factors for these correlations to determine if they supported the null hypothesis or the alternative hypothesis by calculating Bayes Factor test statistic thresholds (null and alternative) using the function *bayesThreshold* from the R package EGCfuncs (v0.1.0, Cunningham, 2025) in alignment with recommendations from Morey and Rouder (2022). With our collected sample size of 174, the calculated null acceptance region is $|r| < 0.17$; and the null rejection region $|r| > 0.24$. Values below $|r| = 0.17$ are used to infer a lack of construct. Values above the null rejection threshold ($|r|$

> 0.24) are tentatively considered support of a possible relationship.

3.1. Planning task results

Table 3 shows the correlations between the tasks intended to measure planning abilities. In alignment with our hypothesis, the correlations between the planning tasks in both samples were largely within the calculated null acceptance region which was pre-registered at $|r| < 0.17$, providing reasonable evidence that these three metrics are not measuring a common underlying construct. There is one exception, which is the correlation between the Tower of London task score and the Traveling Salesperson problems score in the replication sample only. However, while the correlation value in the replication sample exceeded our null acceptance criteria slightly, this was not true in the original sample ($r = 0.15$), suggesting that this result is not stable between samples and would require further investigation via follow up studies (see Supplementary Material Fig. S2 for a graphical representation of planning task relationships). It is further worth noting, as can be seen in Table 3, the Tower of London and Traveling Salesperson tasks are more correlated with other, largely non-overlapping, sets of tasks than they are with each other.

Given the lack of consistent correlations between these tasks, one possible contributing factor may be inadequate internal reliability within the planning measures themselves. In consideration of this possibility, we calculated the split-half reliability of the Tower of London task and the Traveling Salesperson problems (the internal reliability of the Zoo Map test could not be calculated due to the nature of the task being a single problem to solve) and found them to be within the lower bound of expected values ($r = 0.43$ for both, or $r = 0.60$ when corrected via the Spearman-Brown prediction formula as suggested for short tasks using the CTT package in R; Remmers et al., 1927; v2.3.3, Willse, 2018). Further, our values are not inconsistent with the reliability of these tasks in the field in general (e.g., Lemay et al., 2004). Nevertheless, to ensure our correlational results were not due to low task reliability, we calculated additional metrics from each task. Specifically, we calculated time-based metrics which differed from our pre-registered accuracy-based metrics presented above. These alternative outcome measures achieved satisfactory internal split-half reliability in both the Tower of London and the Traveling Salesperson tasks ($r = 0.62$ and $r = 0.85$ respectively, or $r = 0.77$ and $r = 0.92$ respectively when corrected via the Spearman-Brown prediction formula). Yet, even with the more reliable alternative outcome measures, we observed the same minimal correlations between planning measures as in the original findings (see Supplemental Material Table S2 for the correlational results with the alternative metrics extracted from the three planning tasks). Together, these additional analyses suggest that the reliability of the original metrics is not likely to be the primary factor responsible for the lack of relationship found between these tasks.

3.2. Planning and other tasks results

Concerning the correlations of the planning tasks with the other cognitive and personality variables, performance on the Traveling Salesperson task was correlated with performance on the gradual-onset continuous performance task (gradCPT; $r = 0.42$), the speed of processing task ($r = 0.29$), and the spatial working memory task ($r = 0.28$), as well as with one of the measures of creativity ($r = 0.35$) in the original sample and the speed of processing task ($r = 0.25$) in the replication sample. Meanwhile, performance on the Tower of London task was correlated with the gradCPT ($r = 0.36$), and the matrix reasoning task (UCMRT; $r = 0.29$) outcomes in the replication sample. Correlations between the planning measures and tasks measuring adjacent cognitive skills appear to be similarly strong or stronger than any associations found between the planning measures themselves.

Concerning the consistency of correlations between the planning tasks and the other cognitive and personality variables, the only

Table 3
Correlations between Planning Tasks and Other Measures.

Measured Construct (Task Name)	Planning (Tower of London Task)		Planning (Zoo Map Task)		Planning (Traveling Salesperson Task)	
	Original Sample	Replication Sample	Original Sample	Replication Sample	Original Sample	Replication Sample
Planning Tasks:						
Planning (Tower of London)	X	X	0.16	0.15	0.14	0.30
Planning (Zoo Map)			X	X	0.16	0.16
Planning (Traveling Salesperson)					X	X
Other Individual Difference Measures:						
Spatial Reasoning (Paper Folding)	0.10	0.16	0.04	0.03	0.03	0.14
Fluid Intelligence (UCMRT)	0.04	0.29	−0.14	0.20	−0.05	0.16
Spatial WM (Backwards Corsi Block Tapping)	0.17	0.03	0.01	0.07	0.28	0.10
Sustained Attention/Inhibition (gradCPT)	0.14	0.36	−0.16	0.04	0.42	0.17
Speed of Processing (Arrows Response Time)	−0.03	0.20	−0.11	−0.03	0.29	0.25
Multitasking (Predictable Task Switching)	0.05	0.04	−0.14	−0.07	0.09	−0.05
Placekeeping (UNRAVEL - RT)	−0.04	−0.05	0.08	−0.01	−0.04	0.16
Placekeeping (UNRAVEL - accuracy)	−0.32	−0.11	0.02	−0.17	−0.24	0.17
Creativity (Shifting Shapes - Total Shapes)	−0.09	−0.17	0.06	−0.04	0.35	−0.02
Creativity (Shifting Shapes - Unique Shapes)	−0.12	−0.10	0.06	−0.04	0.10	−0.03
Crystallized Intelligence (Shipley Vocabulary)	0.23	0.13	0.20	0.05	0.04	0.12
Extraversion (BFI)	0.15	−0.08	0.24	−0.13	0.06	−0.06
Agreeableness (BFI)	−0.10	0.04	−0.06	0.09	−0.19	0.05
Conscientiousness (BFI)	0.15	0.06	0.13	0.00	−0.06	0.05
Neuroticism (BFI)	0.06	−0.02	−0.09	−0.14	−0.02	−0.21
Openness (BFI)	−0.22	−0.02	−0.08	−0.08	0.26	−0.11

Correlations between the three main planning measurement tasks and all additional cognitive measures and personality characteristic variables collected. See Supplementary Fig. S1 for a graph of the full correlation matrix, including all computed relationships between tasks and surveys.

Note: All correlations are Spearman's rank correlations rounded to the 2nd decimal. Bolded values are greater than the alternative threshold of 0.24. When necessary, scores were reversed to align directionally with other tasks, such that for all metrics, higher values represent better performance.

planning task systematically correlated in both samples is the Traveling Salesperson task with the speed of processing task. The gradCPT showed correlation with the Tower of London in the replication sample but with the Traveling Salesperson in the initial sample, calling for caution. Although a few additional correlations between the planning measures and the other cognitive and personality measures exceeded the alternative hypothesis threshold of $r = 0.24$, these associations were not consistent across samples or planning tasks. This dearth of reliable patterns further underscores the limited convergence among the planning tasks themselves. If these tasks were capturing the same core construct of planning, we would expect to observe consistent associations—both in magnitude and direction—with related constructs such as multitasking, spatial reasoning, or personality traits linked to planning behavior. The fact that such associations do not emerge suggests a lack of cohesion among the planning tasks, or alternatively, a lack of effective measurement when these tasks are administered in isolation.

4. Discussion

The primary research question this study sought to address was whether three planning measurement tasks commonly used to assess planning skills in various fields adjacent to psychology measure a seemingly cohesive construct of planning by associating strongly with one another. In order to answer this question, we performed Spearman's rank correlations to assess the strength of the relationships between the three included planning tasks: the Tower of London task, the Zoo Map Test, and the Traveling Salesperson problems. The results indicated that, although these three tasks are each individually used in the field and reported as measures of planning abilities, they likely do not require utilization of the same cognitive abilities or processes as one another. If these tasks primarily tapped the same underlying skills (in this case we would assume planning would be the most significant skill loaded upon), we would expect to see relatively strong correlations among the three tasks, as well as consistent associations with known adjacent cognitive and personality metrics. This though was not the case. Instead, all correlation values were within our null acceptance region, except for one ($r = 0.30$) that was inconsistent between samples and was far lower

than would typically be associated with the measurement of a truly shared cognitive construct. In sum, the weak and nonexistent correlations between tasks intended to measure planning found in both the original data and, again, in the replication data, point to a lack of common cognitive construct underlying these three tasks.

In evaluating potential explanations for this result, we examined and ruled out both poor internal reliability and ceiling effects as driving factors. Studies on the reliability of these planning tasks are sparse, however, several report similar levels of internal reliability as were found in our data sample (see Lemay et al., 2004 for an overview of several of these studies). For example, Georgiou et al. (2017) found that the split-half reliability for the Tower of London accuracy metric was $r = 0.58$, which is relatively similar to our calculated split-half value for this outcome measure ($r = 0.43$, or $r = 0.60$ when corrected using the Spearman–Brown prediction formula; note that it is unclear whether the reliability values reported by Georgiou et al., 2017 were adjusted using the Spearman–Brown procedure). Other tasks unfortunately have limited internal reliability information available in the literature. However, despite the subpar level of reliability—or lack of confirmed reliability—found in common measures, they remain consistently utilized in the field. Thus, although our goal in the preceding analyses was to assess the cohesiveness of planning as a measured construct, the tasks used to assess this might not be effectively measuring their intended target which raises additional concerns for the field at large. A further concern might be the potential for ceiling effects, especially in the Zoo Map test as this measure is primarily used in clinical populations and with children. However, while many participants showed ceiling level performance, there was a reasonable degree of variability (i.e., participants scoring across the entirety of the scale) on the standard accuracy metric of the Zoo Map task. Even so, because some subset of participants was at or near ceiling-level performance, we also derived an additional metric based on planning time, in line with usage in the literature. This planning-time measure demonstrated sound psychometric properties and showed no evidence of ceiling effects. Moreover, it was strongly correlated with the original accuracy metric, indicating that it captured similar underlying skill variance. At the same time, like the original metric, it exhibited relatively low correlations with time-based metrics

from our other planning tasks, reinforcing our original conclusions about the cohesion between planning tasks (see Supplementary Material Table S2 for correlations between this measure and others in the suite of outcomes).

Apart from measurement reliability issues addressed above, several possible explanations for the results exist. The first is that these particular planning tasks differ in their emphasis on different components of planning. For example, Scholnick et al. (1997) suggested that there are four distinct components of planning: detecting an appropriate order to sequence required actions, noticing the constraints placed on the possible order of those actions, using memory to order the actions while attending to and abiding by constraints, and monitoring the process of action execution in a dynamical problem space. These authors provided an evaluation of two example planning tasks, specifically the Tower of London task and a grocery shopping task, where participants must efficiently gather items from a fictional store diorama by minimizing time and distance. They used these examples to show how different task types emphasize different components of planning abilities. According to their analysis, although these planning tasks share important elements of planning, they also differ substantially in their focus. They asserted that the grocery shopping task places emphasis on attentional control, spatial understanding (or temporal understanding), prior knowledge, and goal search processes, while the Tower of London task primarily places load on management of problem constraints, monitoring, and memory resources. Differing components of planning, or the underlying cognitive abilities necessary to deploy a successful plan, may then be responsible for a lack of visible association between tasks that separately appear to each require planning abilities.

It is also possible that only one of these measures is effectively measuring what researchers might generally agree is the definition of planning skills (Oosterman et al., 2013). One could argue that, for example, the most commonly used task, the Tower of London, is a good candidate for being the “correct” task for measuring planning skills, however, this negates the problem that if this is the case, there still exists no external verification of this claim. Planning is a complex and multifaceted skill and should be able to be measured in multiple ways, as evidenced by the large variety of types of tasks that researchers have employed to assess it. If a single task is assumed to accurately capture the complexity of this skill as it appears in situ, it is more likely that the task only measures a specific component of planning—such as ‘look-ahead’ (Burgess et al., 2005)—rather than the full extent of this complex ability. Indeed, in this study and others, associations between simplistic object displacement tasks and more complex tasks, which arose out of suggestions for better modeling of the true complexities of planning in the real world, have been found to be lacking (Burgess et al., 2005). These results suggest that pure look-ahead tasks and more complex planning tasks may not be loading on the same underlying skills, or if they are, it is to a lesser extent than previously asserted.

A final explanation for the lack of associations found between the three common planning measures included in this study, is that planning as a construct is not cohesive enough to be measured via a single or even a set of measures, or similarly, that planning is more of a semantic category rather than a true cognitive construct as discussed above in the introduction of this paper. If this holds true, then it may be the case that none of the planning measures broadly used are effectively measuring planning skills, and that multiple different types of measures would be necessary to tap into the umbrella category of planning that is commonly referred to semantically. This rationale would provide an explanation for the great breadth of task types and requirements found across literature aiming to assess or understand planning capacity.

Further research is necessary to disentangle these possibilities and point toward a more satisfying strategy for measuring planning abilities or the skills that enable their deployment. Particularly, the identification of consistent components of planning activities could be a fruitful step toward understanding which of these possibilities best explains the lack of cohesion between so-called planning measurement tasks. It may be

the case that planning measurements, and other activities that require the employment of planning skills, can be found to be associated with one another when assessed at a finer grained level. For example, while most planning tasks require planners to assess the implications of the present decision on the future of the problem space, tasks that require the planner to assess the space a greater number of steps into the future (or a long planning horizon) might more strongly associate with one another than they do with tasks that have shorter planning horizons. Similarly, tasks that require planning around interruptions might be more strongly associated with each other than to tasks without interruptions. Indeed, these and other characteristics of planning tasks that can be decomposed from any given activity requiring planning may present a less semantic and more quantitative definition of planning which in turn may be useful in endeavors to first measure, and then intervene on, this important skill.

CRediT authorship contribution statement

Emma G. Cunningham: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Daphné Bavelier:** Writing – review & editing, Conceptualization. **C. Shawn Green:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

Daphne Bavelier is a cofounder of Akili Interactive, Boston, MA. Otherwise, the authors declare that they have no financial, personal, or other relationships with other organizations or companies that may inappropriately impact or influence the research and interpretation of the findings.

Appendix A. Supplementary data

Supplementary material for this article can be found online at <https://doi.org/10.1016/j.cognition.2025.106220>.

Data availability

Data can be accessed via the project OSF site: <https://osf.io/5npvq/>. Analysis scripts available upon request to the authors.

References

- Alderman, N., Burgess, P., Knight, C., & Henman, C. (2003). Ecological validity of a simplified version of the multiple errands shopping test. *Journal of the International Neuropsychological Society*, 9(1), 31–44. <https://doi.org/10.1017/S1355617703910046>
- Altmann, E. M., & Trafton, J. G. (2015). Brief lags in interrupted sequential performance: Evaluating a model and model evaluation method. *International Journal of Human-Computer Studies*, 79, 51–65. <https://doi.org/10.1016/j.ijhcs.2014.12.007>
- Ballhausen, N., Mahy, C. E. V., Hering, A., Voigt, B., Schnitzspahn, K. M., Lagner, P., ... Kliegel, M. (2017). Children's planning performance in the zoo map task (BADS-C): Is it driven by general cognitive ability, executive functioning, or prospection? *Applied Neuropsychology: Child*, 6(2), 138–144. <https://doi.org/10.1080/21622965.2015.1124276>
- Berg, W. K., Byrd, D. L., McNamara, J. P. H., & MacDonald, C. A. (2006). New Indices of Planning Abilities Using the Tower of London Task. In *The Quantified Process Approach to Neuropsychological Assessment*. Psychology Press.
- Burgess, P., Simons, J. S., Coates, L. M.-A., & Channon, S. (2005). The search for specific planning processes. In *The Cognitive Psychology of Planning* (pp. 199–227). London, UK: Psychology Press. <https://books.google.com/books?hl=en&lr=&id>

- =7R15AgAAQBAJ&oi=fnd&pg=PA199&ots=S7ITUyPrJe&sig=ZGw5YpyLbkeFT9QbxGEiAbN4dGw.
- Cai, D., Georgiou, G. K., Wen, M., & Das, J. P. (2016). The role of planning in different mathematical skills. *Journal of Cognitive Psychology*, 28(2), 234–241. <https://doi.org/10.1080/20445911.2015.1103742>
- Cunningham, E. G. (2025). EGCfuncs: Emma G. Cunningham's General Analysis and Visualization Functions (R Package Version 0.1.0). <https://github.com/emmagcunningham/EGCfuncs>.
- Dale, G., Cochrane, A., & Green, C. S. (2021). Individual difference predictors of learning and generalization in perceptual learning. *Attention, Perception, & Psychophysics*, 83(5), 2241–2255. <https://doi.org/10.3758/s13414-021-02268-3>
- Davies, S. P. (2005). Planning and problem solving in well-defined domains. In , Vol. 35. *The Cognitive Psychology of Planning. Current Issues in Thinking & Reasoning*. Psychology Press. <https://books.google.com/books?hl=en&lr=&id=7R15AgAAQBAJ&oi=fnd&pg=PA35&ots=S7IUWSjuH8&sig=fRAHSHeiKMmcOzv9tVya0qFFtM>
- Georgiou, G. K., Li, J., & Das, J. P. (2017). Tower of London: What level of planning does it measure? *Psychological Studies*, 62(3), 261–267. <https://doi.org/10.1007/s12646-017-0416-8>
- Green, C. S., Kattner, F., Eichenbaum, A., Bediou, B., Adams, D. M., Mayer, R. E., & Bavelier, D. (2017). Playing some video games but not others is related to cognitive abilities: a critique of unsworth et al. (2015). *Psychological Science*, 28(5), 679–682. <https://doi.org/10.1177/0956797616644837>
- Hart, Y., Mayo, A. E., Mayo, R., Rozenkrantz, L., Tendler, A., Alon, U., & Noy, L. (2017). Creative foraging: An experimental paradigm for studying exploration and discovery. *PLoS One*, 12(8), Article e0182133. <https://doi.org/10.1371/journal.pone.0182133>
- Hayes-Roth, B. (1980). *Flexibility in Executive Strategies (vol. 1170)*. Rand. <https://apps.dtic.mil/sti/citations/tr/ADA090691>
- Hayes-Roth, B., & Hayes-Roth, F. (1979). A cognitive model of planning*. *Cognitive Science*, 3(4), 275–310. https://doi.org/10.1207/s15516709cog0304_1
- Hering, A., Kliegel, M., Rendell, P. G., Craik, F. I., & Rose, N. S. (2018). Prospective memory is a key predictor of functional independence in older adults. *Journal of the International Neuropsychological Society*, 24(6), 640–645.
- Kessels, R. P. C., van den Berg, E., Ruis, C., & Brands, A. M. A. (2008). The backward span of the Corsi block-tapping task and its association with the WAIS-III digit span. *Assessment*, 15(4), 426–434. <https://doi.org/10.1177/1073191108315611>
- Kliegel, M., McDaniel, M. A., & Einstein, G. O. (2000). Plan formation, retention, and execution in prospective memory: A new approach and age-related effects. *Memory & Cognition*, 28(6), 1041–1049.
- Krikorian, R., Bartok, J., & Gay, N. (1994). Tower of London procedure: A standard method and developmental data. *Journal of Clinical and Experimental Neuropsychology*, 16(6), 840–850. <https://doi.org/10.1080/01688639408402697>
- Lachman, M. E., & Burack, O. R. (1993). Planning and control processes across the life span: An overview. *International Journal of Behavioral Development*, 16(2), 131–143.
- Lemay, S., Bédard, M., Rouleau, I., & Tremblay, P.-L. (2004). Practice effect and test-retest reliability of attentional and executive tests in middle-aged to elderly subjects. *The Clinical Neuropsychologist*, 18(2), 284–302. <https://doi.org/10.1080/13854040490501718>
- Lihoreau, M., Gómez-Moracho, T., & Pasquaretta, C. (2022). Traveling Salesman. In J. Vonk, & T. K. Shackelford (Eds.), *Encyclopedia of animal cognition and behavior* (pp. 7068–7071). Springer International Publishing. https://doi.org/10.1007/978-3-319-55065-7_1821
- Macgregor, J. N., & Ormerod, T. (1996). Human performance on the traveling salesman problem. *Perception & Psychophysics*, 58(4), 527–539. <https://doi.org/10.3758/BF03213088>
- Miotto, E. C., & Morris, R. G. (1998). Virtual planning in patients with frontal lobe lesions. *Cortex*, 34(5), 639–657. [https://doi.org/10.1016/S0010-9452\(08\)70770-6](https://doi.org/10.1016/S0010-9452(08)70770-6)
- Morey, R. D., & Rouder, J. N. (2022). *BayesFactor: Computation of Bayes Factors for Common Designs (Version 0.9.12.4.7) [R Package]*. Comprehensive R Archive Network (CRAN). <https://CRAN.R-project.org/package=BayesFactor>
- Morris, R., Kotitsa, M., & Bramham, J. (2005). Planning in patients with focal brain damage: From simple to complex task performance. In *The Cognitive Psychology of Planning*. Psychology Press.
- Morris, R., & Ward, G. (2005). *The Cognitive Psychology of Planning*. Psychology Press.
- O'Hara, K. P., & Payne, S. J. (1998). The effects of operator implementation cost on planfulness of problem solving and learning. *Cognitive Psychology*, 35(1), 34–70. <https://doi.org/10.1006/cogp.1997.0676>
- Oosterman, J. M., Wijers, M., & Kessels, R. P. C. (2013). Planning or something Else? Examining neuropsychological predictors of zoo map performance. *Applied Neuropsychology: Adult*, 20(2), 103–109. <https://doi.org/10.1080/09084282.2012.670150>
- van Opheusden, B., & Ma, W. J. (2019). Tasks for aligning human and machine planning. *Current Opinion in Behavioral Sciences*, 29, 127–133. <https://doi.org/10.1016/j.cobeha.2019.07.002>
- Pahor, A., Stavropoulos, T., Jaeggi, S. M., & Seitz, A. R. (2019). Validation of a matrix reasoning task for mobile devices. *Behavior Research Methods*, 51(5), 2256–2267. <https://doi.org/10.3758/s13428-018-1152-2>
- Phillips, L. H., MacLeod, M. S., & Kliegel, M. (2005). Adult aging and cognitive planning. *The Cognitive Psychology of Planning*, 111–134.
- Phillips, L. H., Wynn, V. E., McPherson, S., & Gilhooly, K. J. (2001). Mental planning and the tower of London task. *The Quarterly Journal of Experimental Psychology Section A*, 54(2), 579–597. <https://doi.org/10.1080/713755977>
- R Core Team. (2024). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Rammstedt, B., & John, O. P. (2007). Measuring personality in one minute or less: A 10-item short version of the big five inventory in English and German. *Journal of Research in Personality*, 41(1), 203–212. <https://doi.org/10.1016/j.jrp.2006.02.001>
- Remmers, H. H., Shock, N. W., & Kelly, E. L. (1927). An empirical study of the validity of the Spearman-Brown formula as applied to the Purdue rating scale. *Journal of Educational Psychology*, 18(3), 187–195. <https://doi.org/10.1037/h0072665>
- Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, 124, 207–231. <https://doi.org/10.1037/0096-3445.124.2.207>
- Rosenberg, M., Noonan, S., DeGutis, J., & Esterman, M. (2013). Sustaining visual attention in the face of distraction: A novel gradual-onset continuous performance task. *Attention, Perception, & Psychophysics*, 75(3), 426–439. <https://doi.org/10.3758/s13414-012-0413-x>
- Rouder, J. N., Speckman, P. L., Sun, D., Morey, R. D., & Iverson, G. (2009). Bayesian t tests for accepting and rejecting the null hypothesis. *Psychonomic Bulletin & Review*, 16(2), 225–237. <https://doi.org/10.3758/PBR.16.2.225>
- Schmitter-Edgecombe, M., McAlister, C., & Weakley, A. (2012). Naturalistic assessment of everyday functioning in individuals with mild cognitive impairment: The day-out task. *Neuropsychology*, 26(5), 631–641. <https://doi.org/10.1037/a0029352>
- Scholnick, E. K., Friedman, S. L., & Wallner-Allen, K. E. (1997). What do they really measure? A comparative analysis of planning tasks. In *The Developmental Psychology of Planning: Why, How, and When Do We Plan?* (pp. 127–156). Lawrence Erlbaum Associates Publishers.
- Shallice, T. (1982). Specific impairments of planning. *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, 298(1089), 199–209. <https://doi.org/10.1098/rstb.1982.0082>
- Shallice, T., & Burgess, P. W. (1991). Deficits in strategy application following frontal lobe damage in man. *Brain*, 114(2), 727–741. <https://doi.org/10.1093/brain/114.2.727>
- Shepard, R. N., & Feng, C. (1972). A chronometric study of mental paper folding. *Cognitive Psychology*, 3(2), 228–243. [https://doi.org/10.1016/0010-0285\(72\)90005-9](https://doi.org/10.1016/0010-0285(72)90005-9)
- Shipley, W. C. (1940). *Shipley Institute of Living Scale (SILS)*. APA PsycTests.
- Simon, H. A. (1975). The functional equivalence of problem solving skills. *Cognitive Psychology*, 7(2), 268–288. [https://doi.org/10.1016/0010-0285\(75\)90012-2](https://doi.org/10.1016/0010-0285(75)90012-2)
- Spearman, C. (1904). The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1), 72–101. <https://doi.org/10.2307/1412159>
- Vernon, P. A., & Strudensky, S. (1988). Relationships between problem-solving and intelligence. *Intelligence*, 12(4), 435–453. [https://doi.org/10.1016/0160-2896\(88\)90006-2](https://doi.org/10.1016/0160-2896(88)90006-2)
- Vickers, D., Butavicius, M., Lee, M., & Medvedev, A. (2001). Human performance on visually presented traveling salesman problems. *Psychological Research*, 65(1), 34–45. <https://doi.org/10.1007/s004260000031>
- Waldron, S. M., Patrick, J., Howes, A., & Duggan, G. B. (2006). *Planning With Information Access Costs in Mind*.
- Waldron, S. M., Patrick, J., Morgan, P. L., & King, S. (2007). Influencing cognitive strategy by manipulating information access. *The Computer Journal*, 50(6), 694–702. <https://doi.org/10.1093/comjnl/bxm064>
- Willse, J. T. (2018). CTT: Classical Test Theory Functions (R Package Version 2.3.3). <http://CRAN.R-project.org/package=CTT>